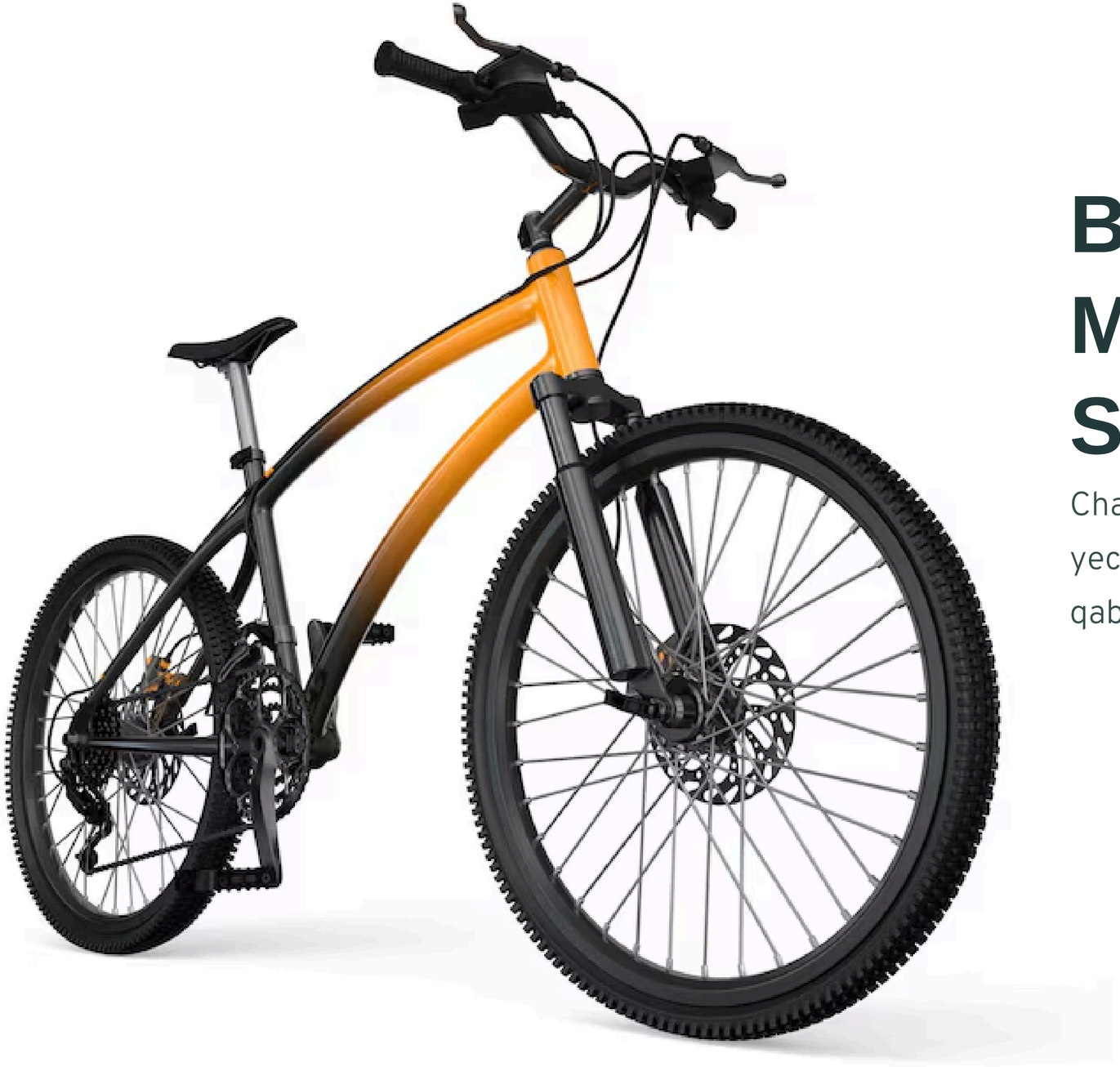


BikeStore 360: Ma'lumotlardan Biznes Strategiyasigacha

Chakana savdoni boshqarish uchun kompleks SQL yechimi — ma'lumot yig‘ish, tozalash, analitika va qaror qabul qilishni avtomatlashtirish.

Ma'lumotlar Bazasi Arxitekturasi

Ikki asosiy sxema: **production** (products, brands, inventory) va **sales** (customers, stores, employees, orders). Sxemalar orasida cheklovlar va FK orqali bog‘lanish — real-time reporting va operatsion izolyatsiya uchun.

1

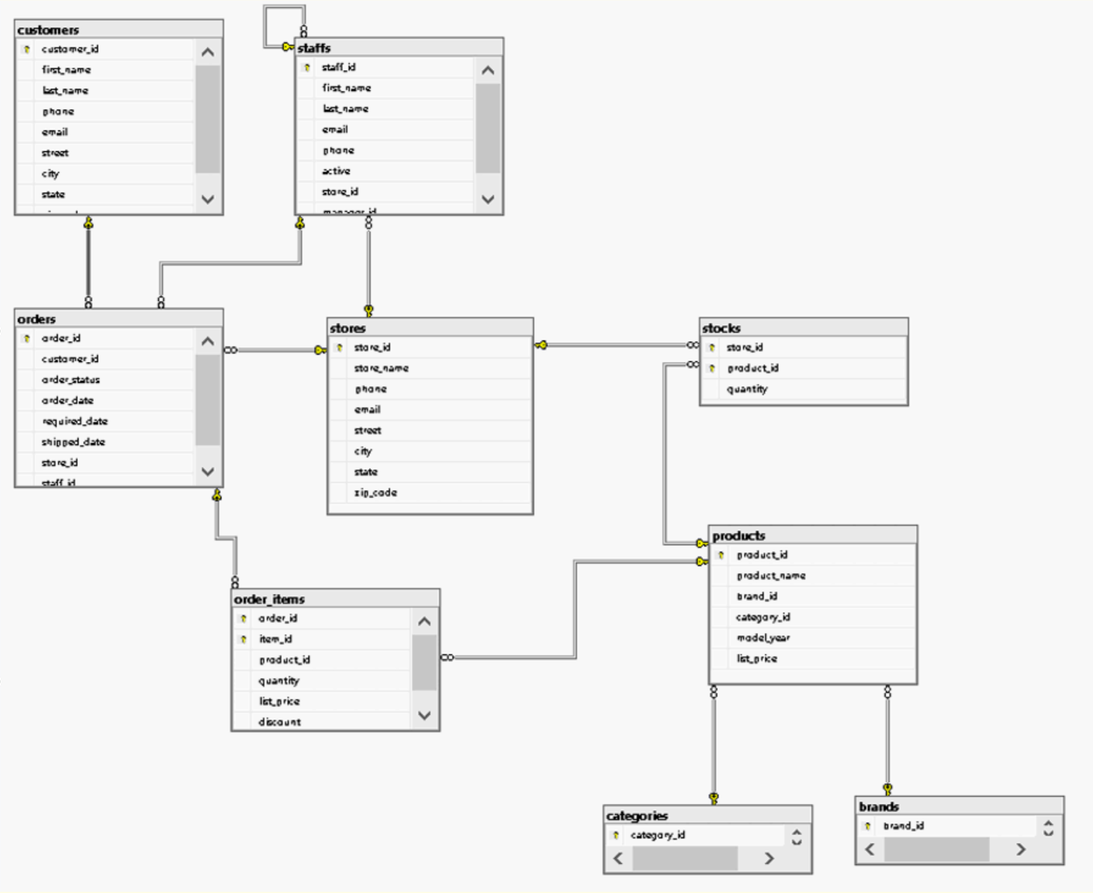
production

Mahsulot katalogi, brend metadatalari, zaxira holati va sotib olish tarixlari.

2

sales

Mijozlar, do'konlar, xodimlar va operatsion buyurtma jadvallari — tahlil uchun optimallashtirilgan.



```sql
--Create table to insert data
CREATE TABLE CSVTest
(ID INT,
FirstName VARCHAR(40),
LastName VARCHAR(40),
BirthDate SMALLDATETIME)
GO
-- Bulk Insert the data from csv file.
BULK INSERT CSVTest
    FROM 'c:\csvtest.txt'
    WITH
    (
        FIELDTERMINATOR = ',',
        ROWTERMINATOR = '\n'
    )
GO
-- Check the data in the table
SELECT *
FROM CSVTest
GO
-- Drop Table for Clean up
DROP TABLE CSVTest
GO
```

Results | Messages

	ID	FirstName	LastName	BirthDate
1	1	James	Smith	1975-01-01 00:00:00
2	2	Meggie	Smith	1979-01-22 00:00:00
3	3	Robert	Smith	2007-11-01 00:00:00

ETL va Ma'lumotlarni Tozalash

CSV fayllar BULK INSERT orqali tez yuklanadi. #orders_stage vaqtinchalik jadvali yordamida:

- Sanalar normalizatsiya qilinadi (TRY_CONVERT/ISDATE filtr).
- NULL qiymatlar uchun biznes qoidalari: default qiymatlar yoki flaglangan yozuvlar.

	product_name	TotalUnitsSold	TotalRevenue	SalesRank
1	Trek Slash 8 27.5 - 2016	154	555558.6111	1
2	Trek Conduit+ - 2016	145	389248.7025	2
3	Trek Fuel EX 8 29 - 2016	143	368472.7294	3
4	Surly Straggler 650b - 2016	151	226765.5510	4
5	Trek Domane SLR 6 Disc - 2017	43	211584.6153	5
6	Surly Straggler - 2016	147	203507.6200	6
7	Trek Remedy 29 Carbon Frameset - 2016	125	203380.8701	7
8	Trek Powerfly 8 FS Plus - 2017	41	188249.6235	8
9	Trek Madone 9.2 - 2017	39	175899.6482	9
10	Trek Silque SLR 8 Women's - 2017	29	174524.7315	10
11	Trek Silque SLR 7 Women's - 2017	30	163079.7282	11
12	Trek Fuel EX 9.8 27.5 Plus - 2017	33	159052.6999	12
13	Electra Townie Original 7D EQ - 2016	290	155169.4138	13
14	Heller Shagamaw Frame - 2016	129	151160.8857	14
15	Trek Fuel EX 9.8 29 - 2017	33	147449.7051	15
16	Electra Townie Original 21D - 2016	289	143393.3928	16
17	Trek Boone 7 - 2017	42	127609.6354	17

Biznes Analitika (Views)

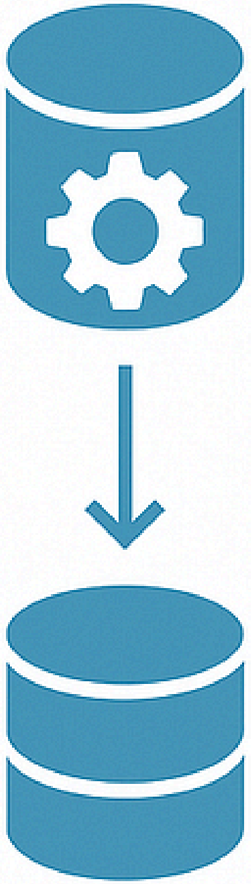
Rejalashtirilgan va tezkor hisobotlar uchun ko‘rinishlar:

vw_StoreSalesSummary

Umumiy daromad, AOV (o'rtacha chek), do'kon va davr bo‘yicha trendlar.

vw_TopSellingProducts

RANK() bilan eng ko‘p sotilgan mahsulotlar, category va brend bo‘yicha.



Stored procedure
Closer to data,
faster, more efficient

Qaror Qabul Qilishni Qo'llab-quvvatlash (Stored Procedures)

Biznes jarayonlarini avtomatlashtirish uchun kalit SPlar:

sp_GenerateRestockList

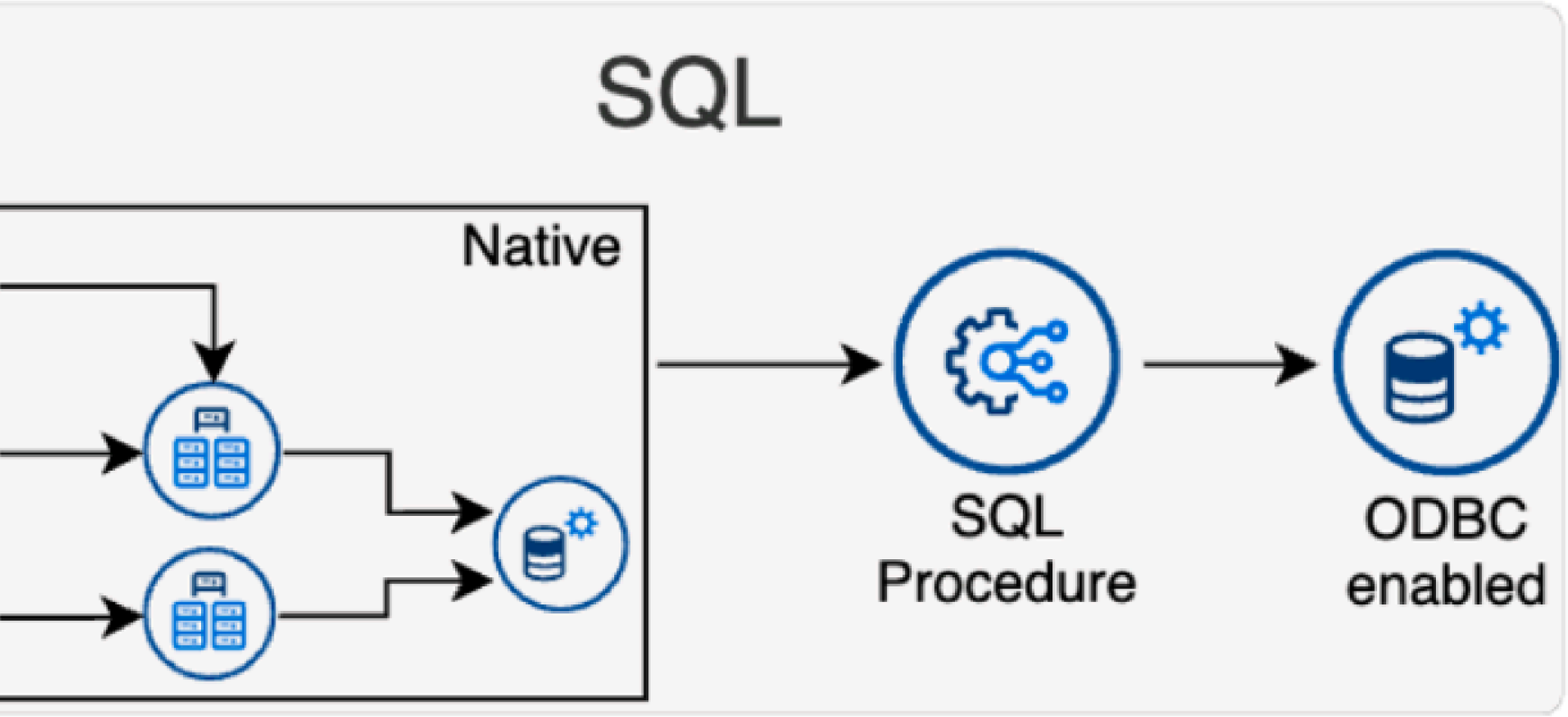
Minimal zaxira va shartlarga asoslangan avtomatik restock ro‘yxati, supplier prioritization bilan.

sp_CompareSalesYearOverYear

Yilma-yil savdo solishtiruvi, foiz o‘sish va segment darajasidagi anomaliyalarni qaytaradi.

sp_GetCustomerProfile

Mijozning barcha xaridlar historiyasi, LTV hisoblash va segmentatsiya atributlari.



Avtomatlashtirish va Monitoring

SQL Server Agent orqali ish oqimi:

- Daily_Sales_Automation_Job — har kuni 01:00 da yuklash, ETL va indeks yangilanishi.
- JobAuditLog — xatolik va ogohlantirishlar (severity, traceback, retry count).
- Alerting — e-mail/Teams integratsiyasi kritik xatolar uchun.

KPI — Asosiy Ko'rsatkichlar

1.2M

Oyga tushum

Jami daromad (oylik hisobot).

42%

Mijozni saqlash

vw_CustomerRetention asosida hisoblangan faol mijozlar foizi.

25%

Top brend hissasi

Eng yuqori 3 brendning umumiy savdodagi ulushi.

Ushbu KPIlar menejerlarga tezkor qaror qabul qilish va kampaniyalarni moslash imkonini beradi.